

Paavani Karuturi

+1 (408)-646-1701 | paavanik1@gmail.com | LinkedIn | GitHub | San Jose, CA

EXPERIENCE

Software Developer Intern

May 2025 - Aug 2025

Olympus

San Jose, CA

- Engineered a production **Python** microservice REST API and instrumented **Prometheus** health metrics, using p95 telemetry to guide tuning that reduced response latency 25% with zero regressions.
- Applied object-oriented design and AI-assisted development tools to iterate on implementation options, then verified code through structured reviews, failure-path debugging, and edge-case tests.
- Documented API behavior and technical tradeoffs while coordinating backend and infrastructure handoffs, translating ambiguous requirements into actionable implementation and release steps.

Software Developer

Dec 2022 - Jul 2024

Fractal Analytics

Mumbai, India

- Delivered 12 production features across **Python**, **TypeScript**, and **Java** for a platform reaching millions of users, carrying requirements through design, implementation, testing, and release.
- Reworked **PostgreSQL** indexes and introduced **Redis** caching on hot read paths, eliminating 60% of redundant computation and reducing query latency by roughly one-third.
- Integrated **Jest** and **Cypress** into **GitHub Actions** CI/CD, raising automated test coverage 40% and reducing production regressions 30% through repeatable UI and API validation.

Software Engineer

Jun 2021 - Dec 2022

Rivach

Hyderabad, India

- Built **Java** REST APIs and **MySQL** data models for backend services serving 25K daily users, resolving concurrency failures under high-frequency writes while preserving data integrity.
- Analyzed query execution plans and tuned database access patterns, reducing p95 latency roughly 30% while maintaining correctness under concurrent load.
- Automated **Docker**-based deployments through **GitHub Actions**, increasing deployment frequency 40% and maintaining zero-downtime releases through consistent delivery practices.

EDUCATION

San Jose State University

San Jose, CA

Master of Science in Software Engineering, GPA: 3.8/4.0

Aug 2024 - May 2026

SKILLS

Languages: Python, Java, C++, TypeScript, JavaScript, Go, SQL

Backend and Frontend: Spring Boot, FastAPI, REST APIs, microservices, React, Next.js, HTML, CSS

Cloud and Infrastructure: AWS, GCP, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD

Data and Quality: PostgreSQL, MySQL, Cassandra, MongoDB, Redis, JUnit, Mockito, Jest, Cypress

AI and Observability: LangGraph, LangChain, GPT-4o, LLM tool use, AI coding assistants, Prometheus, Grafana, OpenTelemetry

Systems and Practices: Data structures, algorithms, object-oriented design, concurrency, distributed systems, debugging, code reviews, documentation

SIDE PROJECTS

Omni-Channel Commerce Platform | *Java, Spring Boot, Kafka, Cassandra, PostgreSQL, Redis, Kubernetes*

- Designed 10 object-oriented **Spring Boot** microservices with 75+ REST APIs and 15 **Kafka** topics, supporting 1M+ products, 500K customers, and 99.95% uptime.
- Load-tested with k6/JMeter to 3,000 concurrent users and 22,000+ requests/min, tuning **Redis** caching to lower p95 latency from 210ms to 38ms.
- Built a 5-agent AI Ops Assistant with 12 **LangGraph/LangChain** tool integrations, then hardened the platform with 400+ tests, 90%+ coverage, 40+ **Prometheus/Grafana** metrics, and CI/CD that cut build time from 20 minutes to 4 minutes.

CERTIFICATIONS & ACHIEVEMENTS

AWS Certified Solutions Architect - Associate | Google Cloud Computing Foundations | AWS Academy Graduate

Runner-up - Stanford LLM x Law Hackathon (2025) | Winner - Cloud Native Hackathon (2022) | Google Cloud: Generative AI with Vertex AI